

Urban Pulse - Smart Tourism & Cultural Intelligence Platform

STONE 2 - GEOSPATIAL, WEATHER & TRANSIT ANALYTICS

TEAM - 1

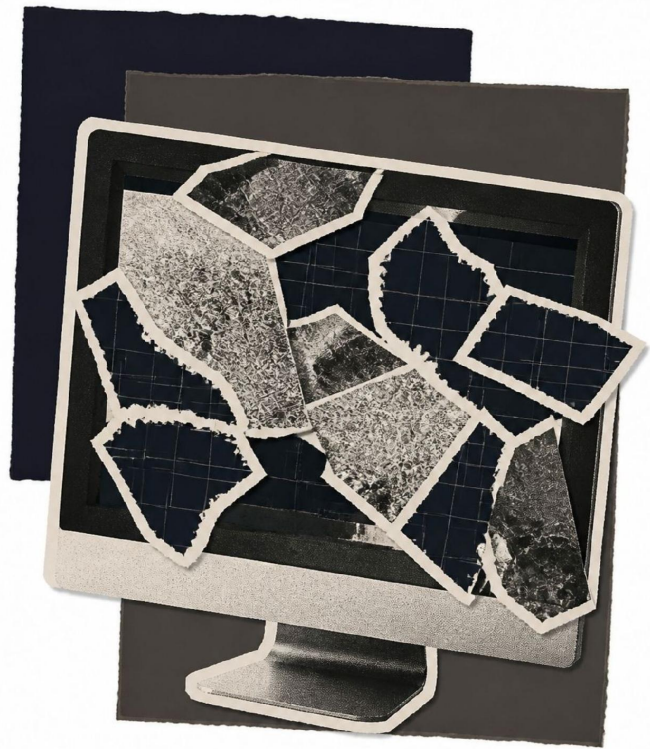
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Problem Statement

In Milestone 1, we established that India's tourism data is fragmented across government sources, lacking actionable visibility. A clean schema alone doesn't help tourism boards make strategic decisions.

Milestone 2 addresses this: we turn our Galaxy schema and real government data into interactive, geospatial, and climate-aware dashboards - making **overcrowding, underutilization, and weather-driven demand shifts** visible and actionable.

Problem Objectives

- 1

Spatial Demand Visualization

Visualize spatial tourist demand concentration across states, heritage sites, and monuments.
- 2

Site Performance Tracking

Track site performance: capacity vs. actual footfall, funding efficiency, and accessibility.
- 3

Multi-Level Drill-Down

Enable seamless multi-level drill-down analysis from national to state to individual site level.
- 4

Weather-Driven Insights

Reveal how weather patterns (temperature, rainfall, humidity, AQI) drive footfall shifts at outdoor heritage destinations.
- 5

Dynamic Dashboards

Transition from static tables to dynamic, filter-driven intelligence dashboards.

Project Scope

In Scope

- Footfall density heatmaps across states/monuments
- Origin-destination flow maps (foreign source country to destination state)
- Site performance dashboards (capacity, congestion, budget efficiency)
- 3-level geographic drill-down
- Weather & Climate overlays (Temperature, Rainfall, Humidity, AQI) via Looker Studio & Open-Meteo
- Temporal filters (Year, Quarter, Month, Peak vs. Off-Peak)

Out of Scope

- Predictive forecasting models (*Milestone 3*)
- Anomaly detection (*Milestone 3*)
- Final executive accessibility index (*Milestone 4*)

Proposed Solution

Our Milestone 2 platform spans **four modular, connected intelligence components** built on a shared data foundation:



Demand Heatmaps & Tourist Flow Maps

Spatial concentration mapping and origin-destination arc flows.



Site Performance & Reliability Dashboards

Capacity, congestion, and funding efficiency tracking.



Geographic Drill-Down Analysis

National to state to monument-level navigational filtering.



Weather Overlays & Temporal Filters

Dual-axis weather/AQI vs. footfall analytics with dynamic parameter controls.

Technology Stack



Supabase

Cloud SQL database hosting the Galaxy Schema and custom aggregated views (`view_weather_tourism_analysis`).

Looker Studio

Internal analytical engine utilized for prototyping spatial bubble maps, validating weather parameter filters, and testing preliminary chart configurations.

Streamlit

Icore interactive platform providing web-based dynamic dashboarding and custom parameter controls

Open-Meteo API

External weather data pipeline supplying historical monthly metrics (Temp, Rainfall, Humidity, AQI).

- ❏ Team uses a blended stack across specialized modules, targeted for unified integration in the final milestone.

About Our Datasets



fact_attractions
&
fact_monthly_tou
rism

Monument footfall,
festival funding, and
monthly Foreign
Tourist Arrivals
(FTAs).



dim_weather /
Open-Meteo API

Historical monthly
weather parameters
(Temperature,
Rainfall, Humidity,
AQI).



dim_country

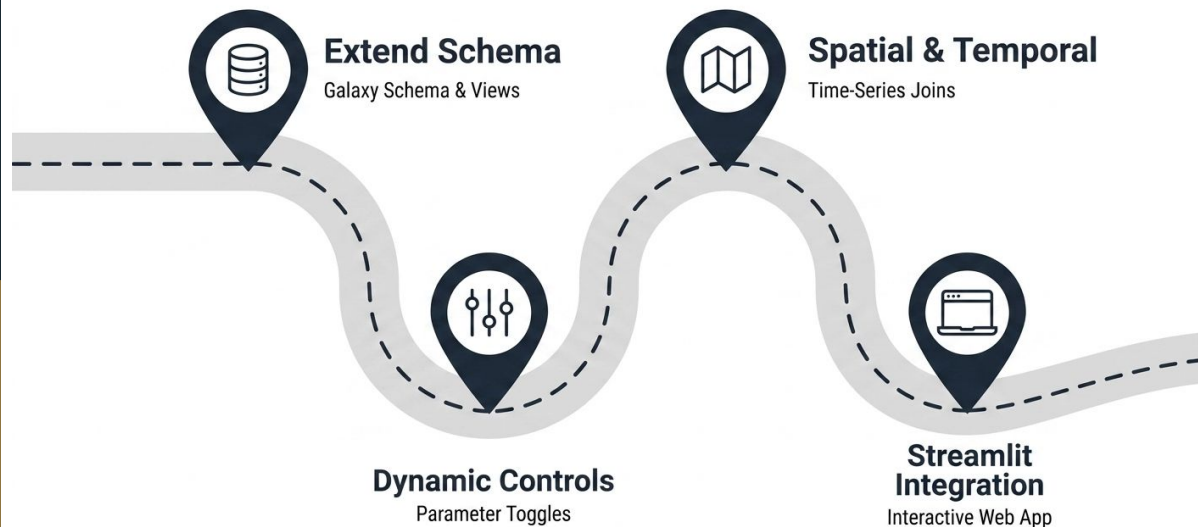
Foreign tourist
source country
mapping for
origin-destination
flow analysis.



dim_State &
dim_Location

Regional hierarchy
supporting
geographic
drill-down.

Methodology



Key Implementation Details

Dynamic Parameter Controls (`p_weather_metric`)

Allows seamless metric toggling on a single combo chart without cluttering views.

Dual Calculated Fields

Numeric value field for scaling chart axes alongside formatted string fields with correct units (°C, mm, %, AQI) for hover tooltips.

Spatial & Temporal Joining

Time-series joins between `Dim_Date` and weather data to isolate seasonal patterns.

Galaxy Schema Extension

Built upon the Milestone 1 Galaxy Schema by adding `dim_country` for origin-destination mapping and joining `fact_monthly_tourism` with `dim_weather` via custom SQL views.

Expected Outcomes



Interactive Heatmaps

Pinpointing high-density tourist clusters and foreign tourist corridors.



Operational Dashboards

Flagging over-capacity vs. underutilized heritage sites.



3-Tier Drill-Down

Functional geographic view: National
→ State → Monument.



Climate Impact Proof

Data-backed evidence of Q2/Q3 monsoon surges driving drop-offs in tourist arrivals.



Scalable Pipeline

Fully ready for Milestone 3 predictive forecasting.

Conclusion



Milestone 2 successfully converts our raw data foundation into **actionable, interactive intelligence**. By combining spatial analysis, performance metrics, and weather correlations, such as the inverse relationship between monsoon peak rainfall and foreign arrivals, we demonstrate that **climate factors directly shape tourism demand**.

This sets a solid foundation for predictive modeling in Milestone 3.

Milestone 1

Galaxy Schema & Clean Data Foundation

Milestone 2

Geospatial, Weather & Interactive Dashboards

Milestone 3

Predictive Forecasting & Anomaly Detection